

Atomic Absorption, Methods and Instrumentation

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Introduction

Atomic absorption spectroscopy has become one of the most frequently used tools in analytical chemistry. This is because for the determination of most metals and metalloids the technique offers sufficient sensitivity for many applications and is relatively interference free. There are two basic atom cells (a means of converting the sample, usually a liquid, into free atoms) used in atomic absorption spectroscopy: (1) the flame and (2) the electrothermal heating of a sample cell. It is generally acknowledged that if sufficient analyte is present in the sample, then it should be determined using a flame technique because this has added advantages of being rapid (assuming only a few elements need be determined) and, in comparison with alternative techniques, very simple to use. Electrothermal atomic absorption spectroscopy (ETAAS) or electrothermal vaporization atomic absorption spectroscopy (ETVAAS) requires more operator skill and is less rapid, but yields substantially superior limits of detection when compared with flame atomic absorption spectroscopy (FAAS). This section describes some of the methods and instrumentation that have been developed for both flame and electrothermal techniques of atomic absorption spectroscopy.

Instrumentation

The basic principle of both FAAS and ETAAS is that a sample is introduced into the atom cell, where it is desolvated and then atomized. The analyte atoms so formed then quantitatively absorb light in a way that is proportional to the concentration of the atoms of the analyte in the cell. The light, which is at a specific wavelength, is then isolated from other wavelengths that may be emitted by the atom cell and then detected. Thus, much of the instrumentation used for ETAAS and FAAS is identical. Both techniques require a similar light source, background correction system, line isolation device (monochromator or polychromator), detector (photomultiplier or charge-coupled device), and readout system. Each of these components is discussed in the following sections, together with details of the individual atom cells (flame and the electrothermal atomizer) and sample introduction systems.

Light Source

The fundamental requirement of the light source is to provide a narrow line profile with little background. It should also have a stable and reproducible output with sufficient intensity to ensure that a high signal-to-noise ratio is obtained. Two basic types of light source are used for atomic absorption, of which the hollow-cathode lamp (HCL) is the more commonly used. This is a lamp in which the cathode is coated with the analyte metal of interest. Within the lamp, inert filler gas (neon or argon) is ionized by an electric current and these ions are then attracted by the cathode. The inert gas ions bombard the cathode and in so doing excite the metal ions coated on it. It is this excitation of the metal that produces the emission of radiation with wavelengths characteristic of the analyte. HCLs are available for most metallic elements. A schematic diagram of a HCL is shown in Figure 1.

Electrodeless discharge lamps are used less frequently than the HCLs except for analytes such as arsenic and selenium. These lamps may be excited using either microwave energy (although these tend to be less stable) or radiofrequency energy. The radiofrequency-excited lamps are less intense than the microwave-excited ones, but are still 5–100 times more intense than a standard HCL. In the electrodeless discharge lamp, a bulb contains the element of interest (or one of its salts) in an argon

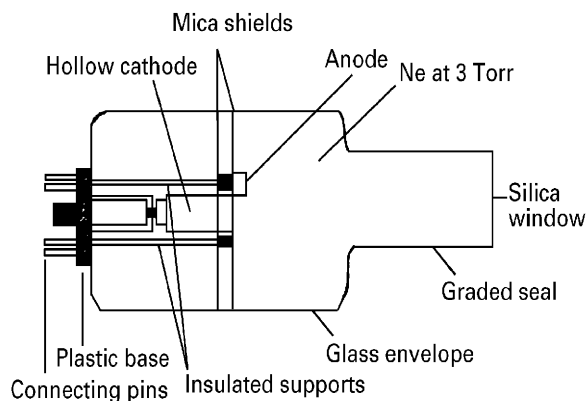


Figure 1 Schematic representation of a hollow-cathode lamp. Reproduced from Ebdon L (1998) *Introduction to Analytical Atomic Spectrometry*, with permission from Wiley.

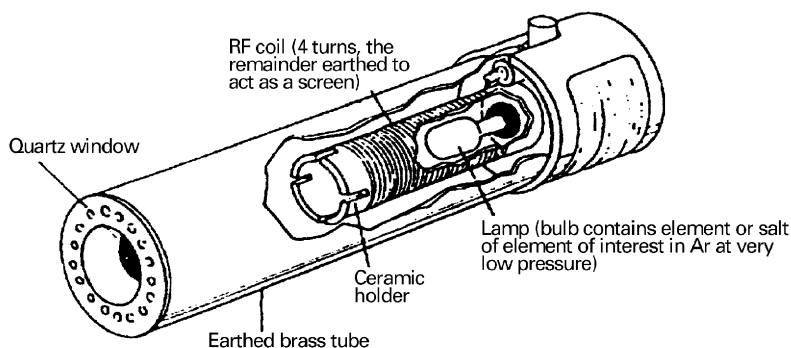


Figure 2 Electrodeless discharge lamp. Reproduced from Ebdon L (1998) *Introduction to Analytical Atomic Spectrometry*, with permission from Wiley.

atmosphere. The radiofrequency energy ionizes the argon and this, in turn, excites the analyte element, causing it to produce its characteristic spectrum. For analytes such as arsenic and selenium, these lamps give a better signal-to-noise ratio than HCLs and have a longer useful lifetime. A schematic diagram of an electrodeless discharge lamp is shown in Figure 2.

Background Correction Systems

There are basically three types of automatic background correction system available for atomic absorption, although manual methods such as the use of nearby nonatomic absorbing lines to estimate background absorbance may also be used. The three main automatic methods are the deuterium or hydrogen lamp, the Zeeman effect, and Smith–Hieftje background correction. The deuterium lamp produces a continuum of radiation, some of which will be absorbed by molecular species within the atom cell. The amount of atomic absorption observed using the deuterium lamp is negligible and hence the atomic absorption signal is obtained by subtracting the absorbance of the continuum lamp from the total analyte absorbance of the HCL. The deuterium or hydrogen lamp is of most value at wavelengths in the UV region ($< 350\text{ nm}$). The Zeeman effect background correction system is more versatile. It relies on a strong magnetic field operating at approximately 1 T and $50\text{--}60\text{ Hz}$, placed either around the light source or, more commonly, around the atom cell to split the signal into a number of components. The π component is at the normal analyte wavelength; the σ components are typically of 0.01 nm (depending on the strength of the magnetic field) on either side of the π component and hence lie outside the atomic absorption profile. Background can be corrected by subtracting the absorbance with the magnetic field ‘on’, from the signal with the magnet ‘off’. It should be noted that this is a simplified description of the process, as different elements have different splitting patterns and the magnetic field may be applied

longitudinally or transversely, which may also produce different splitting patterns.

When utilizing the Smith–Hieftje system, the HCL is boosted periodically to a much higher current, causing the lamp to ‘self-absorb’. In this state no atomic absorption occurs in the atom cell but the molecular absorption still remains. Background correction is achieved automatically by subtracting the signal obtained at high current from that obtained using the normal current. The process is particularly efficient at removing interferences such as that caused by phosphate or selenium determinations, although the high currents used may shorten the lifetime of the lamp. In modern instruments, modulation of the source is achieved electronically. This enables discrimination between absorption and emission signals. Previously, a rotating sector, often referred to as a ‘chopper’, placed between the source and the atom cell was used.

Single-Beam and Double-Beam Instruments

The vast majority of instruments used for atomic absorption measurements have a single-beam configuration, using the optical layout shown in Figure 3a. The double-beam arrangement (Figure 3b) is far more complex and offers far fewer advantages for atomic absorption when compared with double-beam spectrometers in molecular spectroscopy. This is because the reference beam does not pass through the atom cell. Despite this, double-beam instruments can compensate for source drift and for warm-up and source noise.

Line Isolation Devices

To ensure that only light of a wavelength specific to the analyte of interest is being measured, a line isolation device is required. Until recently, the line isolation device used for atomic absorption was a monochromator. There are numerous types of monochromator, but

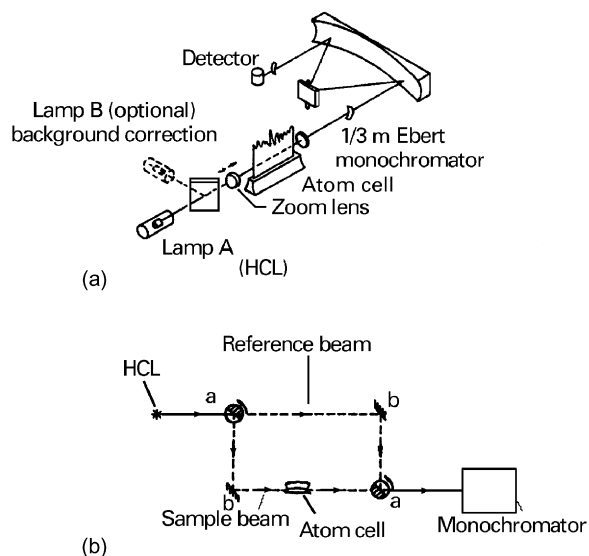


Figure 3 Schematics of (a) single-beam and (b) double-beam spectrometers. Reproduced from Ebdon L (1998) *Introduction to Analytical Atomic Spectrometry*, with permission from Wiley.

modern instruments typically use one of three designs: Ebert, Czerny–Turner or Littrow configuration. These are shown schematically in Figure 4.

Light of all wavelengths enters the monochromator through an entrance slit and is then split into specific wavelengths using either a prism, or more commonly, a diffraction grating. By altering the position of this dispersing element, light of only the desired wavelength passes through the exit slit to the detector. Since the method of atomic absorption is so specific, very highly resolving monochromators are not required. Thus, the focal length of an atomic absorption monochromator is often 0.25 or 0.5 m compared with a minimum of 0.75 m required for conventional optical emission spectroscopy.

A monochromator enables only one wavelength to be interrogated at any instant; this was something of a weakness of atomic absorption spectroscopy. New technology, however, has enabled the development of multi-element spectrometers that use a bank of between 4 and 6 HCLs and an echelle-style polychromator employing orders of 100 or more and are thus capable of considerable dispersion. In addition, such instruments have no exit slit, and so a huge number of wavelengths may be focused onto an array detector such as a charge-coupled device. A schematic diagram of such an instrument is shown in Figure 5. Research continues into producing a continuum light source that would enable 30–40 analytes to be determined simultaneously.

Detection Systems

Traditionally, detection of the light isolated by the monochromator has been accomplished using a

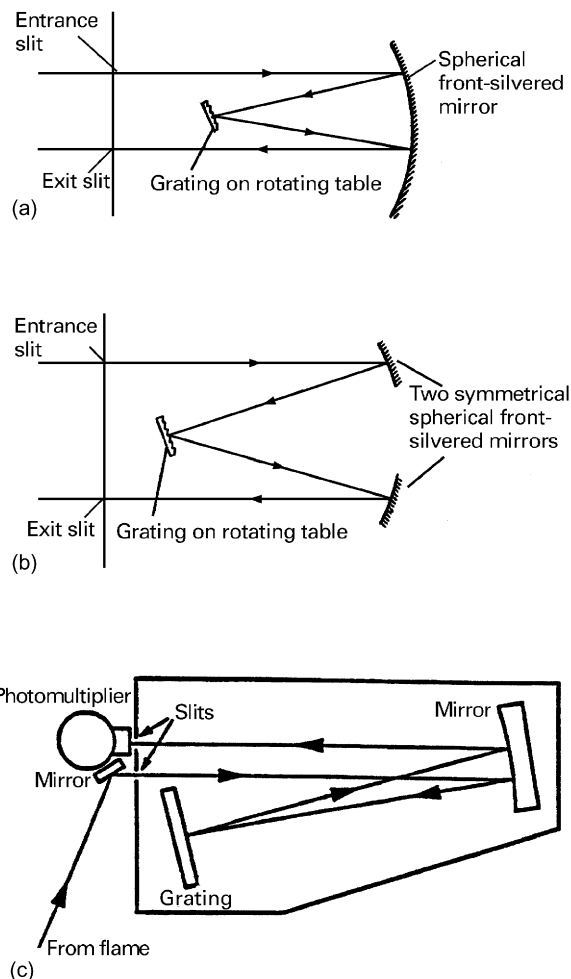


Figure 4 Schematics of different monochromator types: (a) Ebert, (b) Czerny–Turner, (c) Littrow. Reproduced from Ebdon L (1998) *Introduction to Analytical Atomic Spectrometry*, with permission from Wiley.

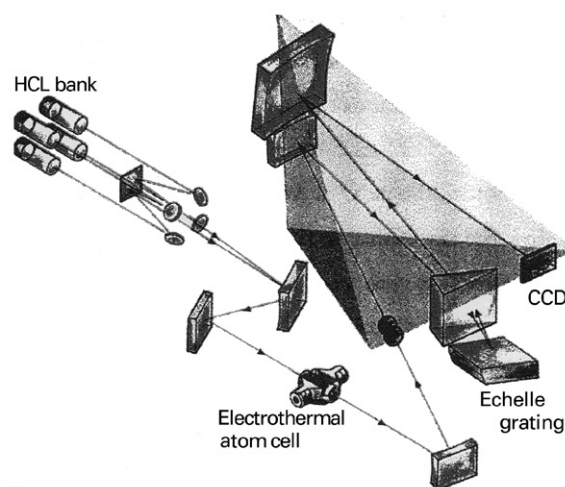


Figure 5 Echelle-based polychromator.

photomultiplier tube (PMT). Several configurations exist, for example end-on and side-on, and the construction may be of different materials, to increase the efficiency at different wavelengths; but basically they all work in a similar way. Light enters the multiplier through a quartz window and impacts with a photocathode that is usually made from one of a number of alloys (e.g. Cs-Sb, Na-K-Sb-Cs or Ga-As), which then emits electrons. These electrons are then accelerated down a series of dynodes, each being at a more positive potential than the previous one. As the electrons impact with successive dynodes, further electrons are ejected and hence a cascade effect occurs. In this way a single photon may cause the ejection of 10^6 electrons. The number of electrons is then measured at the anode and the resulting current is proportional to the radiation reaching the PMT.

As described previously, a number of different materials may be used to coat the photocathode in the PMT, and each has a different response curve. Some, for example the Cs-Sb tube, have very little sensitivity above 600 nm, but others, for example Ga-As, may be used up to almost 900 nm. The response profiles of some commonly used photomultipliers are shown in Figure 6.

Conventional spectrometers use a detector that is capable of measuring only one signal. Multielement instruments use state-of-the-art diode arrays or charge-coupled devices that can measure numerous spatially

separated signals and can therefore simultaneously determine the signals arising from a bank of HCLs. A charge-coupled device may be considered to be similar to an electronic photographic plate. The device consists of several hundred linear photodetector arrays on a silicon chip with dimensions of typically 13×18 mm. The line isolation device (often an echelle grating) separates the analytical wavelengths, and these may then be detected on different regions of the array. A detailed description of how the array works is beyond the scope of this text.

Instrument Control and Output Devices

Manual controls and dials have slowly disappeared, with the large majority of modern instruments being controlled by computer. In addition to controlling the instrumental parameters, the computer may be used to programme autosamplers, save experimental parameters and data, calibrate with known standards, plot calibration curves, use scale expansion facilities (if necessary) and produce statistical data from the results. A continuous graphics mode enables the signal to be monitored over a period of time. This is especially useful when flow injection, chromatography or hydride-generation transients are to be detected and integrated. Older instrumentation relied on a chart recorder for the same purpose.

Sample Introduction and Atom Cells

For flame systems, the sample is introduced via a nebulizer and spray chamber assembly. Sample is drawn up the intake capillary by the Venturi effect. The liquid sample column comes into contact with the fast moving flame gases and is shattered into small droplets. In many systems it is further smashed into still smaller droplets by an impact bead. The flame gases then carry the aerosol (the nebular) through a spray chamber containing a series of baffles or flow spoilers. These act as a droplet size filter, passing the larger droplets to waste (85–90% of the sample) and allowing the finer droplets to be transported to the flame. For electrothermal atomizers, the sample is placed into the atom cell either using a hand-held micropipette or by an autosampler. Since most instruments are supplied with an autosampler and the precision associated with their use is superior to that of the micropipette, most laboratories make use of this method of sample introduction.

As described previously, there are two basic types of atom cell—flame and electrothermal tube—but both can have numerous modifications. The flame cell most commonly used is the 10 cm length air–acetylene burner. This provides a flame that is at approximately 2300 °C

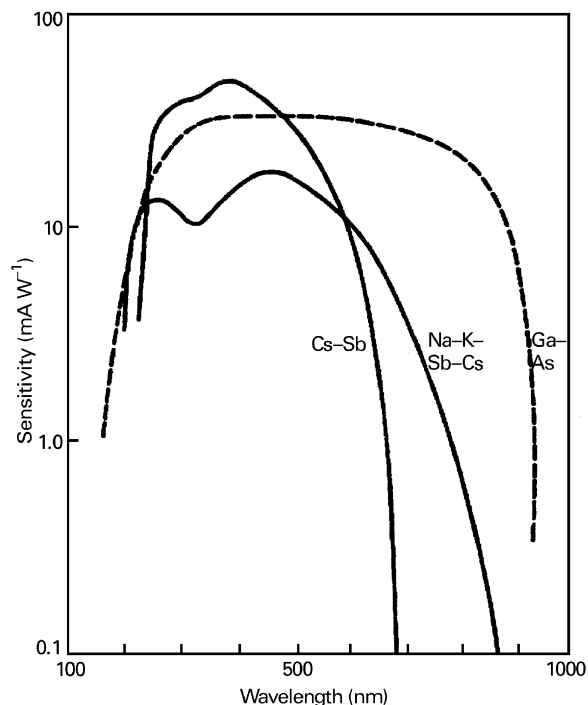


Figure 6 Response curves for several commercial photomultiplier tubes. Reproduced from Ebdon L (1998) *Introduction to Analytical Atomic Spectrometry*, with permission from Wiley.

(although the actual temperature will depend on the fuel/air ratio). The flame may be used in a fuel-lean mode, which produces a hot, oxidizing blue flame; in a fuel-rich mode, which produces a cooler, reducing yellow flame; or in a stoichiometric mode whose properties are between the two. Different analytes give different sensitivities depending on the flame type. Chromium, for instance, gives better sensitivity in a reducing flame, whereas for magnesium it is better to use a lean, oxidizing flame.

Different areas of the flame have different temperatures and different chemical properties. The efficiency of atomization of analytes will therefore depend critically on the area within the flame. Hence, it is important that the light beam from the HCL passes through the region of the flame where atomization is optimal. Optimization of the burner height is therefore necessary to ensure that maximum sensitivity is achieved.

Although an air-acetylene flame is sufficient for the atomization of the majority of analytes, it is not sufficiently hot or reducing to atomize analytes that form refractory oxides (e.g. Al, Mo, Si and Ti). For analytes such as these, a nitrous oxide-acetylene flame of length 5 cm is used. This flame is hotter (2900 °C) and contains typically 2.5–3.0 times as much fuel as the air-acetylene flame. Other flame types, for example air-propane and air-hydrogen also exist, although they are less commonly used. The air-hydrogen flame (2200 °C) is almost invisible and has the advantage of being transparent at lower wavelengths offering improved noise characteristics for elements such as lead or tin at around 220 nm. The inhomogeneity of flame chemistry and the need for burner height optimization hold true for all flame types.

One common modification to a flame cell is to place a quartz tube on the burner head so that the light beam passes through the length of the tube. This method is especially useful when using sample introduction by hydride generation or by gas chromatography. The tube is heated either by a flame or electrically using a wire winding. The analyte, which enters the tube via a hole or slit cut into the side, is atomized within the tube. The atoms then leave the tube but, because of their retention in the tube, spend longer in the light beam than in normal flame systems. An increase in sensitivity is therefore obtained. Another small modification of this system is that, instead of a quartz tube, a quartz T-piece may be used. However, the function is the same.

Electrothermal atom cells have changed radically since their inception in the late 1950s. The majority of electrothermal devices have been based on graphite tubes that are heated electrically (resistively) from either end. Modifications such as the West Rod Atomizer (a carbon filament) were also devised but were later abandoned. Tubes and filaments made from highly refractory metals such as tungsten and tantalum have also been made, but

they tend to become brittle and distorted after extended use and have poor resistance to some acids. Their use continues, however, in some laboratories that need to determine carbide-forming elements. For example, silicon reacts with the graphite tube to form silicon carbide, which is both very refractory and very stable. The silicon is therefore not atomized and is lost analytically. Use of a metal vaporizer prevents this.

In electrothermal atomization the sample is introduced into the tube, which is then heated in a series of steps at increasing temperature. The sample is dried at a temperature just above the boiling point of the solvent (but not so hot as to cause frothing and spitting of the sample); ashed (charred at an intermediate temperature to remove as much of the concomitant matrix and potential interferences as possible without losing any analyte); and then atomized at a high temperature. During the atomization stage, the atoms leave the graphite surface and enter the light beam, where they absorb the incident radiation. The ashing and atomizing temperatures used will depend on the analyte of interest; for example, some analytes such as lead are relatively volatile and so cannot be ashed at temperatures above 450 °C, otherwise volatile salts such as chlorides will be lost. Other analytes, for example, magnesium, are less volatile and can be ashed at temperatures close to 1000 °C without analyte loss. Such elements require a much higher atomization temperature. The sensitivity of ETAAS is greater than that for FAAS because the atoms are formed within the confines of a tube and hence spend longer time in the light beam. Also, since the sample is placed within the atom cell, 100% of it is available for analysis compared with the 10–15% available in flame systems. A comparison of characteristic concentrations (concentration that gives an absorbance of 0.0044) obtained for flame and electrothermal techniques for many analytes is shown in [Table 1](#). It must be stressed that the figures for ETAAS will depend critically on the injection volume and so values tend to be given in absolute terms (i.e. a weight).

Overall, the Massmann design of electrothermal atomizer in which the tube is heated from either end is still the most common ([Figure 7b](#)), but more recently transversely heated tubes have been developed ([Figure 7a](#)). The longitudinally heated tubes have a temperature gradient along the tube, with the central portion being several hundred degrees hotter than the ends. This can lead to condensation of analyte at the cooler ends and subsequent reatomization from the hot graphite surface. Several atomization peaks may therefore result. The transversely heated tubes do not have a temperature gradient and therefore do not suffer from this problem.

Table 1 Comparison of characteristic concentrations for FAAS and ETAAS

Analyte	FAAS ($\mu\text{g L}^{-1}$)	ETAAS (pg)
Ag	30	5
Al	300	30
As	800 ^a , 5 ^b	42
Au	100	18
B	8 500	600
Ba	200	15
Be	16	2.5
Bi	200	67
Ca	13	1
Cd	11	1
Co	50	17
Cr	50	7
Cs	40	10
Cu	40	17
Fe	45	12
Ge	1 300	25
Hg	2 200 ^a , 0.1 ^b	220
K	10	2
La	48 000	7 400
Mg	3	0.4
Mn	20	6
Mo	280	12
Ni	50	20
Pb	100	30
Rb	30	10
Sb	300	60
Se	350 ^a , 4 ^b	45
Si	1 500	120
Sn	400	100
Sr	40	4
Te	300	50
Ti	1 400	70
Tl	300	50
U	110 000	40 000
V	750	42
Yb	700	3
Zn	10	1

^aUnder normal flame conditions.^bWith vapour generation.

Interferences

Flame techniques are regarded as being relatively free from interferences, but some distinct classes of interference do exist. These include a few spectral interferences (e.g. Eu at 324.753 nm on Cu at 324.754 nm), ionization interferences and chemical interferences. Ionization interference is a vapour-phase interference (in the past often termed cation enhancement) that occurs when the sample contains large amounts of an easily ionized element. The presence of large concentrations of easily ionized elements will lead to a large concentration of electrons in the flame. These electrons prevent the ionization of the analyte and hence lead to higher atomic absorption signals. If the easily ionized element is not present in the standards, the analyte may be partially ionized (and lost analytically) and hence serious

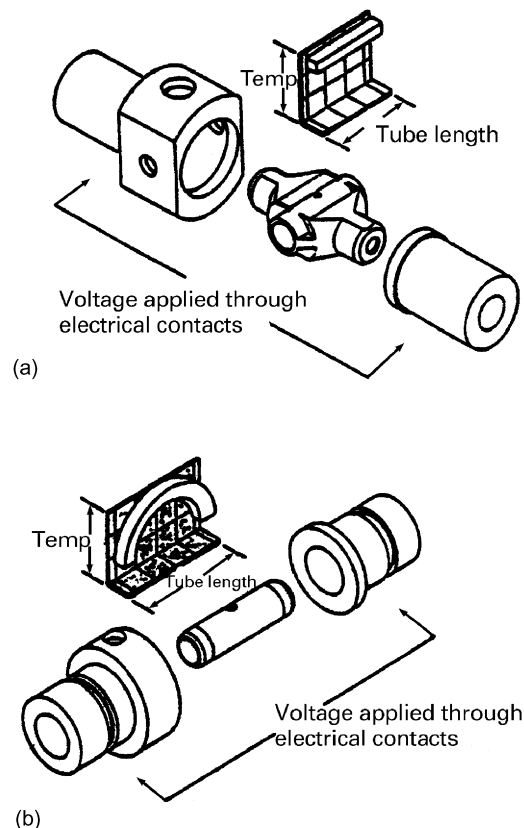


Figure 7 Electrothermal tubes available commercially: (a) transversely heated graphite atomizer (THGA), (b) longitudinally heated Massmann atomizer. Reproduced from Ebdon L (1998) *Introduction to Analytical Atomic Spectrometry*, with permission from Wiley.

overestimates of the true concentration will be obtained. This type of interference may be overcome by adding an excess of easily ionized element to all standards and samples.

Chemical interferences may exist in several different forms:

- Formation of less volatile compounds, for example when phosphate is present during the determination of calcium. Calcium phosphate is refractory and hence atomization will be retarded in comparison with calcium in the standards.
- Formation of more volatile compounds, for example chlorides.
- Occlusion into refractory compounds. Small amounts of analyte may become trapped in a refractory substance and hence not be atomized efficiently.
- Occlusion into volatile compounds. Some compounds sublime explosively and hence atomization may be enhanced.

Most of these interferences may be overcome by using a hotter flame (e.g. nitrous oxide–acetylene); by adding a

chelating reagent (e.g. EDTA) to complex preferentially with the analyte; by adding a releasing agent (e.g. lanthanum) that will combine preferentially with phosphate; or by optimizing the flame conditions and viewing height.

ETAAS was renowned for being highly prone to interferences. However, modern methods and instrumentation have decreased this problem substantially. Interferences include memory effects, chemical interferences (loss of analyte as a volatile salt, carbide formation, condensation and recombination), background absorption (smoke) and physical interferences such as those resulting from placing the sample on a different part of the tube. The stabilized temperature platform furnace (STPF) concept has gone a long way in eliminating the problem. The concept is that a 'matrix modifier' is required for interference-free determinations. This is a material that either decreases the volatility of the analyte, enabling higher ash temperatures to be achieved and hence boiling away more interference (a common example is a mix of palladium and magnesium nitrates) or increases the volatility of the matrix (for example, the introduction of air 'burns away' many interferences leaving the analytes in the atomizer). The matrix modifier may be placed in the autosampler and added to all samples and standards. Other requirements for the STPF concept include a rapid heating rate during atomization, integrated signals (rather than peak height), a powerful background correction system (e.g. Zeeman), fast electronics to measure the transient signal, and isothermal operation. Isothermal operation means that the analyte is vaporized from the graphite surface into hot gas. In normal ETAAS, the analyte leaves the hot tube wall and enters the cooler gas phase. Under these circumstances it may recombine with some other species to form a compound and thus be lost analytically. To overcome this, platforms that are only loosely in contact with the tube walls have been developed. The analytes in this case are atomized not by the resistively heated graphite tube but by the surrounding gas. The transversely heated tubes described earlier have a platform as an integral part of the tube. A modification to this technique, called probe atomization, has also been developed. Here the sample is placed on a probe and is then dried and ashed in the normal way. The probe is then removed from the tube, which is heated to the atomization temperature. The probe is then reintroduced into the hot environment.

Most analytical techniques for use with furnace work now utilize the advantages of the STPF concept. Other interferences, for example formation of carbides, may be partially overcome by treating the atomizer with a carbide-forming element, such as by soaking in tantalum solution. The principle is that the tantalum occupies the active sites on the surface, thereby preventing the analyte from forming a carbide.

Methods

Numerous methods have been described for use with AAS, but the majority require the sample to be in a liquid form. Some ETAAS systems allow the analysis of solids directly (e.g. by weighing small amounts onto sampling boats that may be slotted into specialized tubes), but usually, if solids are to be analysed, acid decomposition methods are required. An alternative is the analysis of slurries, using finely ground material ($<10\ \mu\text{m}$) dispersed in a solvent. Manual agitation of the slurry ensures homogeneity of the sample, enabling a representative aliquot to be introduced. Some autosamplers come equipped with an ultrasonic agitator that will perform the same task. This method of analysis is frequently used in ETAAS.

If samples containing high levels of dissolved (or suspended) solids are to be analysed by FAAS, a flow injection technique (in which discrete aliquots of sample are introduced) may be used. This has also been referred to as 'gulp sampling'. This prevents salting up of the nebulizer and blockage of the burner slot, which are obviously undesirable effects that have an adverse effect on sensitivity and signal stability. Flow injection techniques may also be used to preconcentrate analytes. If the sample flows through a column containing an ion-exchange resin, the analytes will be retained. Elution with a small volume of acid may yield very high preconcentration factors. This technique has been used for both FAAS and, more recently, for ETAAS, yielding limits of detection far superior to those achieved under normal conditions. A typical flow injection manifold suitable for this type of application is shown in [Figure 8](#).

A more traditional method of preconcentration is solvent extraction. FAAS may be used as a detector for analytes in organic solvents (sensitivity may even be improved because of improved nebulization efficiency), but occasional blocking of the burner slot by carbon may occur. This may be removed by gentle rubbing with a noncombustible material (such as a spatula). Organic solvents may also be introduced to ETAAS, but the 'dry'-stage temperature has to be modified to prevent sample loss by spitting.

Hydride generation is a common method for the detection of metalloids such as As, Bi, Ge, Pb, Sb, Se, Sn and Te, although other vapours, for example Hg or

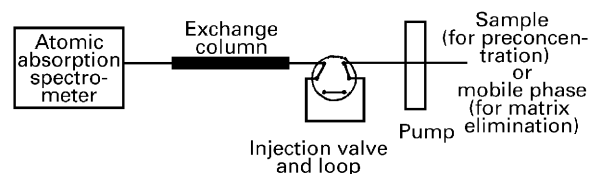


Figure 8 Typical flow injection manifold for matrix analyte preconcentration or matrix elimination.

alkylated Cd, may also be determined. This technique improves the sensitivity of the analysis substantially. Since the sample is in the gas phase, the sample transport efficiency is close to 100%. The hydrides atomize readily in the flame, although this approach is usually used in conjunction with a quartz T-piece in the atom cell. Methods have been developed that trap the hydrides on the surface of a graphite tube for use with ETAAS. This leads to preconcentration and further improvements in detection limit.

Chromatography has been coupled to AAS to effect speciation analysis. Here different chemical forms of an analyte are separated (either by high-performance liquid chromatography (HPLC) or, if they are sufficiently volatile, by gas chromatography) prior to introduction to the atomic absorption instrument. Gas chromatography is usually coupled directly with a T-piece interface arrangement, but HPLC couplings introduce the sample through the conventional nebulizer/spray chamber assembly. A small postcolumn air bleed may be necessary to compensate for the differences in flow rate between the chromatograph ($1\text{--}2\text{ ml min}^{-1}$) and the uptake rate of the nebulizer ($5\text{--}10\text{ ml min}^{-1}$). Chromatography is not frequently coupled with ETAAS because the atom cell is not well suited to continuous monitoring.

The limited linear range offered by the AAS technique may be partially overcome by using alternative, less sensitive lines. This is a useful technique that avoids the requirement to dilute samples. Care must be taken, however, to ensure that viscosity effects do not cause a difference in nebulization efficiency between samples and standards. Alternatively, burner head rotation may be used. This shortens the path in the flame through which the light beam travels and hence decreases sensitivity. However, there is often an increase in noise as a result of using this approach.

Conclusions

The sensitivity offered by the various techniques utilizing AAS for the determination of metals and metalloids is often sufficient for many applications. Direct aspiration of samples in solution into a flame atomic absorption

instrument provides the analyst with rapid acquisition of data with good precision. The technique is remarkably free from interferences and those that do exist may usually be overcome by judicious choice of operating conditions. The instrumentation involved is relatively simple and, for flame work, analyses may easily be performed by nonexpert operators. If increased sensitivity is required, the analyst has the option of preconcentrating the analyte using one of a number of methods prior to introduction to FAAS, or alternatively may use ETAAS. The latter technique, however, does require more experience if reliable results are to be obtained.

A potential disadvantage of using atomic absorption is that in the past instruments have been capable of determining only one analyte at a time. Modern instrumentation has improved this number to as many as six, although it must be stressed that compromise conditions (e.g. in ETAAS temperature programmes) must be used. This may lead to a decrease in overall sensitivity compared with single-element determinations. However, the ongoing development of multielement AAS will certainly boost the use of this already popular and versatile technique.

See also: Atomic Absorption, Theory, Atomic Spectroscopy, Historical Perspective, Light Sources and Optics.

Further Reading

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